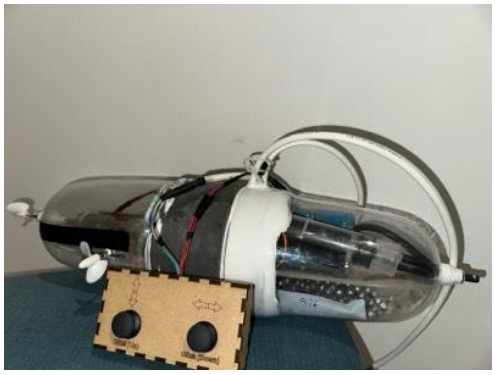
Your exploring buddy Nemo
(Submarine)

Photos of the robot (fully assembled)



Description of all the parts needed for construction

Technical:

1x Arduino (I used Arduino mega)

2x DC Motors (4v - 8v) – one for forward and back movement and one for yaw

1x Motor Driver (I used TB6612) – to control the movement of the motors

1x Servo Motor (I used MG995 360') – to control the buoyancy of the boat

1x Infra-Red Sensor – to measure the distance from the submarine to the floor, if the distance is less than 10cm it won't allow you to go further down (this function was commented for testing purposes)

1x Water Sensor – for security if it detects water in the submarine it will automatically rise up ignoring your commands

1x Breadboard – it will make your life easier (I cut it up into small pieces so it would fit)

2x Joystick (I used module KY-023 PS2 JOYSTIK DUAL-AXIS) – to control the movements of the submarine

Battery pack (3×AA ≈ 4.5V)
Battery pack (5xAA ≈ 7.5V)

https://alfaelektronik.com.mk/index.php
https://gacelektronik.com.mk/

Other:
2x Propeller – for movement
2x Flat Bases – to have a ground for every part
2x Plastic vases (the body)
2x PVC pipes – to connect the body
1x Syringe 60ml – to control the buoyancy
1x Tube – to get the water in the syringe
1x Gear and Rack – to move the syringe
Weights – to control the weight compared to the volume

Connections:

TB6612 STBY → Arduino PWM 7

TB6612 PWMA → Arduino PWM 2

TB6612 AIN1 → Arduino PWM 8

TB6612 AIN2 → Arduino PWM 9

TB6612 PWMB → Arduino PWM 3

TB6612 BIN1 → Arduino PWM 10

TB6612 BIN2 → Arduino PWM 11

TB6612 gnd → breadboard 28d

TB6612 vm → breadboard 27d

TB6612 vcc → breadboard +

TB6612 AO1, AO2 → motor1(forward/back) +,-

TB6612 BO1, BO2 → motor2(yaw) +,-

Battery pack (3×AA ≈ 4.5V) positive → breadboard 27e

Battery pack (3×AA ≈ 4.5V) negative → breadboard 28e

Servo red → breadboard 27c

Servo brown → breadboard 28b

Servo orange → breadboard 24a

Water Sensor red(+) → breadboard +

Water Sensor brown(-) → breadboard –

Water Sensor orange(S) → Arduino Digital 30

IR Sensor red → breadboard +

IR Sensor black → breadboard –

IR Sensor yellow → Arduino analog in A2

Arduino 5v → breadboard +

Arduino gnd → breadboard –

Arduino pmw 5 → breadboard 24b

Arduino gnd (from pmw) → breadboard 28c

Joystick1 +5v → breadboard +

Joystick1 gnd → breadboard –

Joystick1 VRy → Arduino A0

Joystick1 sw → Arduino Digital 32

Joystick2 +5v → breadboard +

Joystick2 gnd → breadboard –

Joystick2 VRx → Arduino A1

Joystick2 sw → Arduino Digital 31

Battery pack (5×AA ≈ 7.5V) powers the Arduino mega

Description of the construction method, time required, difficulties you encountered during construction and how you solved them

I started the submarine by doing research on what do I need for the main functionalities, firstly I wanted to make it wireless with antennas but soon I read that I won't be too efficient since it's easy to loose signal when it's under water so I decided to make it wired and using a controller.
Knowing that I started gathering all the parts I thought I needed, some I didn't need and some I missed, that was one of my first challenges knowing what I need to complete the functionalities I wanted.

I started to play around with the Arduino to test how it works with the ultra-sonic sensor that I wanted to use to measure the distance of the floor. I soon dropped the idea since the sound would bounce from the walls of the body so I got the infra-red sensor instead.
as I wanted to connect the DC motors that I got from an old rc toy boat (that's also where I got the propellers from) I had no idea how to make them to spin clockwise and counter clockwise, that's where I found out that I need to use a motor driver.

At this point I had everything I needed I made the initial code that wasn't working completely correct, but I had connected the Arduino, servo, motor driver and motors. The Arduino at this point was connected to my laptop and the motors and the servo on a separate battery pack with 3x AA batteries.
YouTube video of the first test:
https://youtube.com/shorts/eO71fE2Xg78?feature=share

After that test I connected the IR sensor and the water sensor and changed the code so everything worked perfectly.

Now all I had to do was assemble the brain of my submarine with the body, but first I need to make the body. This is where I thought "okay I've done similar stuff before, this will be easy". It wasn't.
First I had to find a way to connect my cylinders together so I decided to use PVC pipes and silicone to connect both front and back of the submarine. That included a lot of patience, melting and precision. In the end though it ended up just fine. After that I did the base of both sides and laser cut the rack and the gear for the syringe and servo to be connected.
I made a base for the servo and syringe so the syringe won't move around when the servo rotates, let's call that the "buoyancy device". Initially the buoyancy device had one micro servo SG90 which wasn't powerful enough to move the syringe so I added two. Now the two SG90 could move it but they couldn't move the beginning, so I ended up buying a new and more powerful servo MG995 and that worked perfectly.
YouTube video on how was the progression of the buoyancy device:
Test 1:https://youtu.be/KvesIV3IpE4
Test 2: https://youtube.com/shorts/U92DGamgzfY?feature=share
Test 3: https://youtube.com/shorts/VqYGFaYlNlY?feature=share

With the motors I didn't have much problem besides that I needed to add oil for them to rotate more freely.
In the end I did the calculations on how much weight I need to add to the submarine to have the perfect buoyancy for me to control it with the syringe.
All this may sound simple but I can freely say that it wasn't since I've made many different wire configurations and size adjustments so that everything can fit within the body, in addition being aware of potential water leakage.
Regarding all the other obligations that I had it took about 4-7 weeks where most of the time I used for research at the beginning.

Description of the purpose of the robot

The submarines purpose is to allow the user of the submarine to explore the waters

link to a video of using the robot uploaded to YouTube showing all possible actions of the robot

https://youtube.com/shorts/cHT0ld34t0A?feature=share

Source code

https://github.com/Engjell02/Nemo-submarine

Plans for future improvement of the robot

I would love for the submarine to have higher endurance and hopefully add a camera of some sort to have live feed to explore deeper under water. Maybe even mass production.

References:

https://youtu.be/tGEZApd9MMk?si=0pgqKtTXWaCKTN27
https://youtu.be/KLEH8RJsYgI?si=1l8q54dricrrE5My